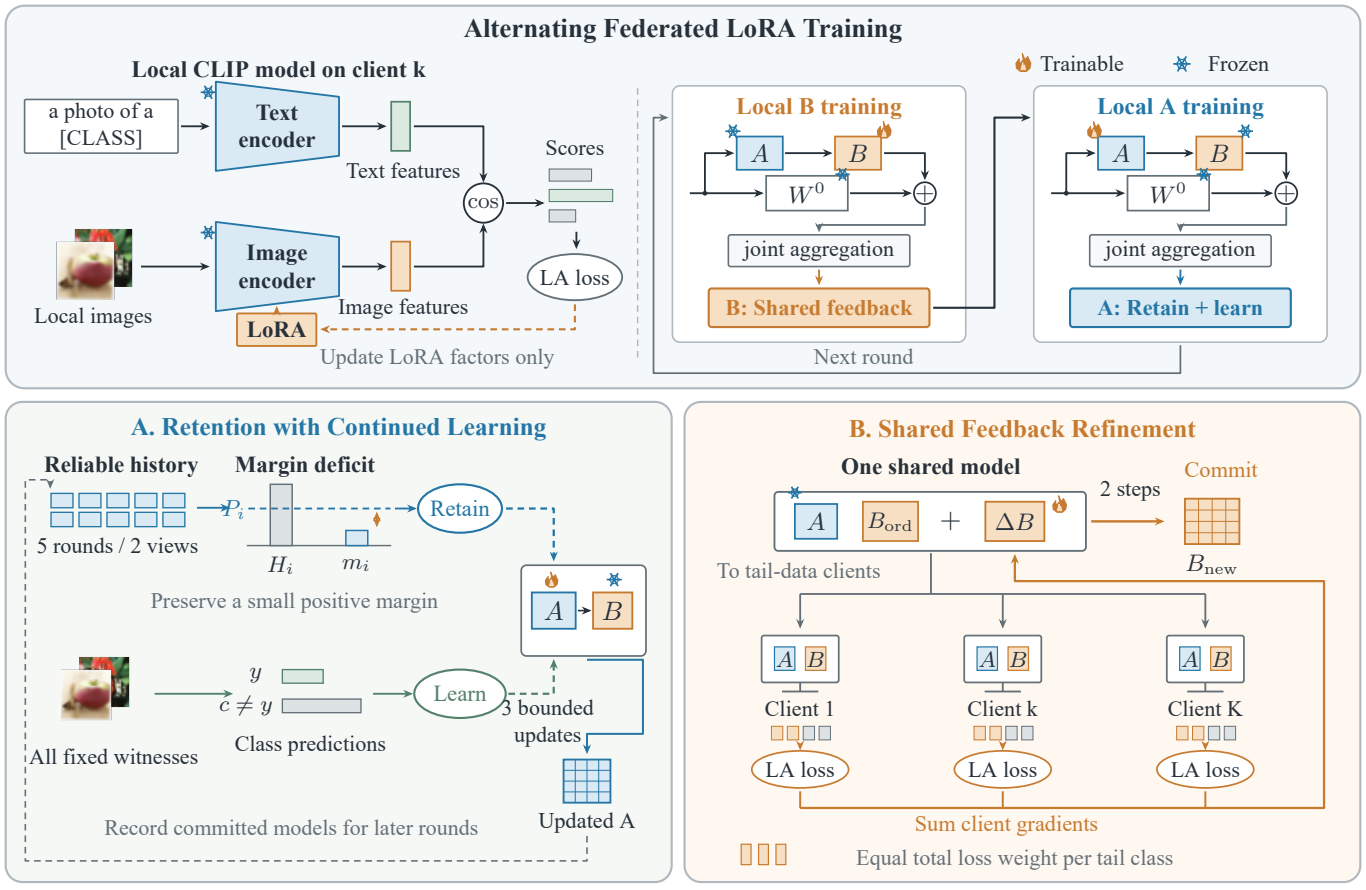

Alternating Federated LoRA Training
Local CLIP model on client k
a photo of a [CLASS]
Text encoder
Text features
Scores
cos
Image encoder
Image features
LA loss
Local images
LoRA
Update LoRA factors only
Trainable
Frozen
Local B training
joint aggregation
B: Shared feedback
Local A training
joint aggregation
A: Retain + learn
Next round
A. Retention with Continued Learning
Reliable history
Margin deficit
5 rounds / 2 views
Retain
Preserve a small positive margin
All fixed witnesses
Class predictions
Learn
3 bounded updates
Updated A
Record committed models for later rounds
B. Shared Feedback Refinement
One shared model
2 steps
Commit
To tail-data clients
Client 1
Client k
Client K
LA loss
Sum client gradients
Equal total loss weight per tail class